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Christian College
ACC No.

324352(24)

B. E. (Third Semester) Examination, April-May., 2016

(New Scheme)

(Elect. Engg. Branch)

ELECTRICAL MACHINES–I

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Attempt part (a) of every question and any two parts of (b), (c) and (d).*

(a) What is ideal transformer? 2

(b) Define voltage regulation of a transformer. For which

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type of load the voltage regulation is negative? Derive the expression using the equivalent circuit.

(c) Explain open circuit and short circuit tests of a single phase transformer giving circuit diagram for each test. Also mention uses of these tests. 7

(d) A 220/400 V, 10 kVA, 50 Hz single phase transformer has at full load, a copper loss of 120 W. If it has an efficiency of 98% at full load, unit power factor determine the iron losses. What would be the efficiency of the transformer at half full load at 0·8 p.f. lagging. 7

2. (a) What is distribution transformer? 2

(b) Explain Back to Back test on transformer. How can it be used for measurement of efficiency? 7

(c) Explain excitation phenomenon in transformers. 7

(d) A 400/100 V, 5 kVA, two winding transformer is to be used as an auto-transformer to supply power at 400 V from 500 V source. Draw the connection diagram and determine the kVA output of the auto-transformer. 7

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3. (a) What is an open-delta system? 2

(b) Explain with the help of connection and phasor diagrams, how Scott connections are used to obtain two-phase supply from 3-phase supply mains. 7

(c) Give the equivalent circuit and application of 3 winding transformer. Explain how the parameters can be determined experimentaly. 7

(d) Describe the four phasor groups pertaining to 3-phase transformers. Draw the connection schemes and phasor diagrams for each of these four groups. 7

4. (a) Name the various parts of a d.c. machine. 2

(b) Explain the methods of improving commutation with relevant figures. 7

(c) What is the critical field resistance of d. c. shunt generator? What is its significance? 7

(d) A shunt generator gives full load output of 30 kW at a terminal voltage of 200 V. The armature and shunt resistances are 0·05 Ω and 50 Ω respectively. The iron and friction losses are 1000 watt. Calculate generated e.m.f. and efficiency. 7

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5. (a) What is the necessity of a starter for a d.c. motor? 2

(b) Discuss different methods of speed control of a d.c. motors. 7

(c) Describe Swinburne's test with the help of a neat diagram to find out the efficiency of a d.c. machine. 7

(d) A 220 V d.c. series motor is running at a speed of 800 r. p. m. and draws 100 A. Calcualte at what speed the motor will run when developing half the torque. Total resistance of armature and field is 0·1 Ω. Assume that the magnetic circuit is unsaturated. 7

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